PulseRun

Computer Graphics Project - Checkpoint 1: Technical Foundation

Evidence of graphical output (pure C++ renderer)

Group members

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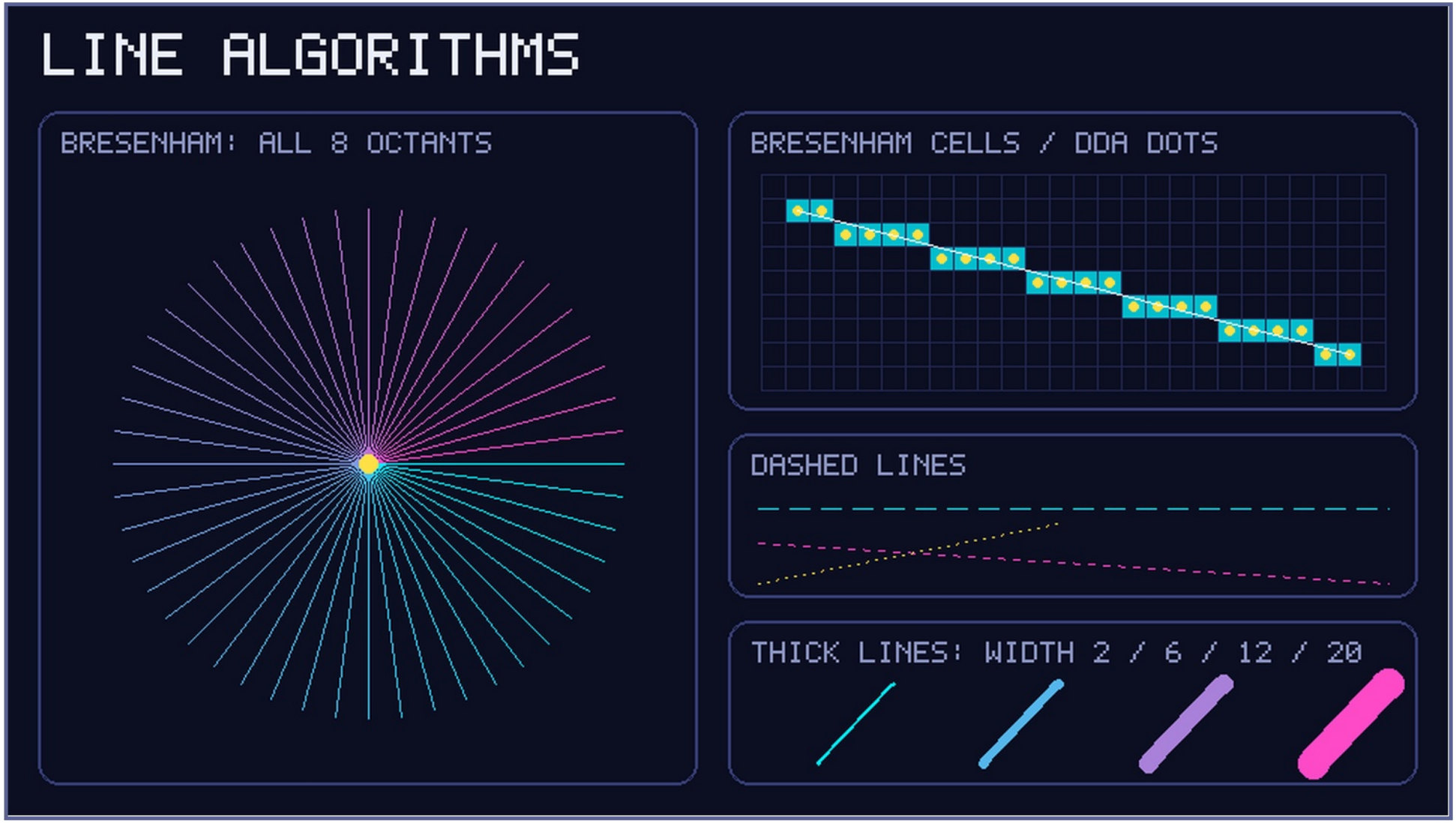
Mathenge Gitahi

Diana Mucheke

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github.com/MathengeGit/pulserun

Line algorithms



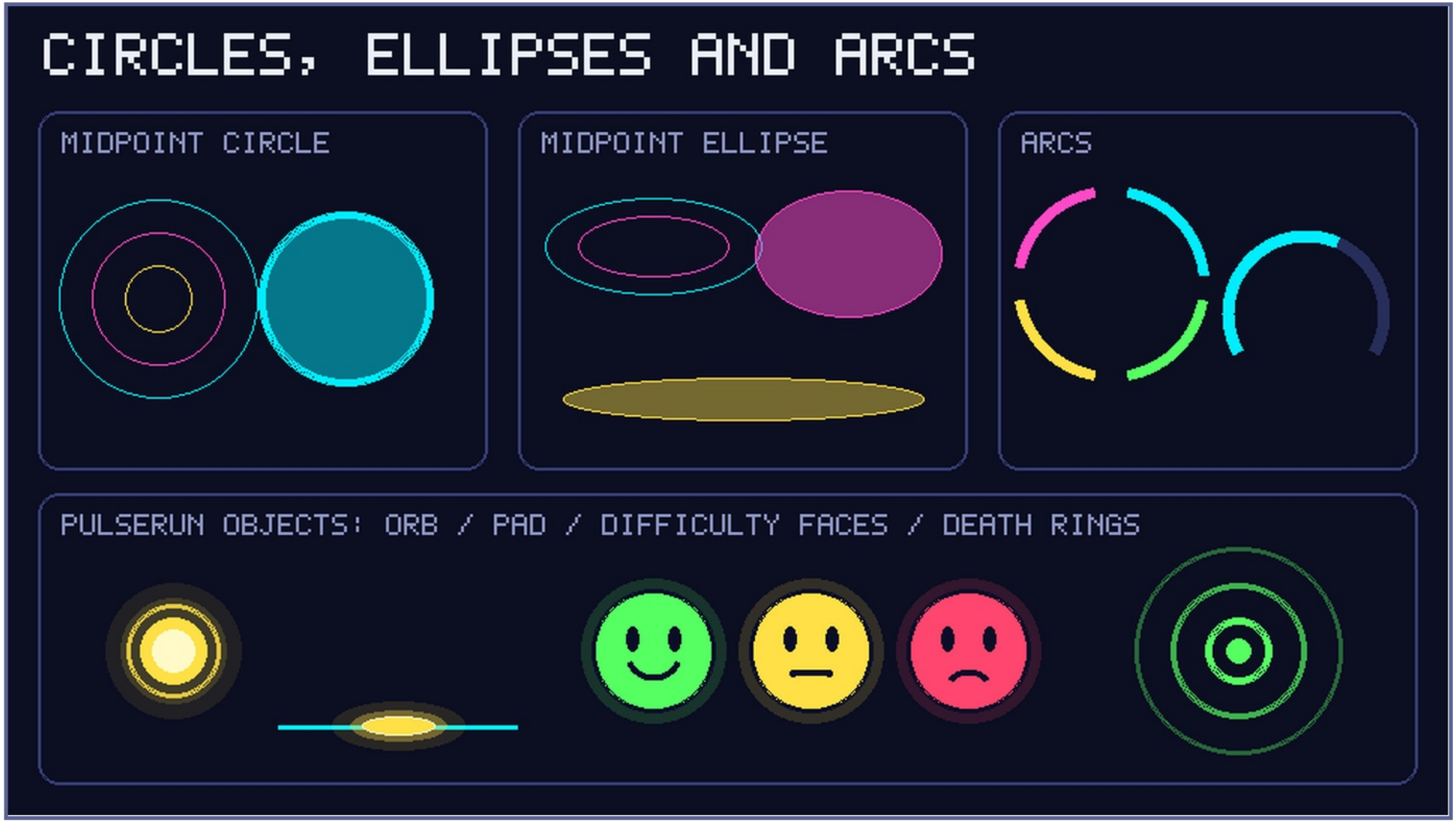
DDA, Bresenham (all 8 octants), dashed lines and thick lines.

Source: src/raster/line.cpp, demos/demo_lines.cpp

Used in PulseRun for: ground line, scrolling grid ticks, saw spokes, UI frames, jump-trail segments.

01_lines.png

Circles, ellipses and arcs



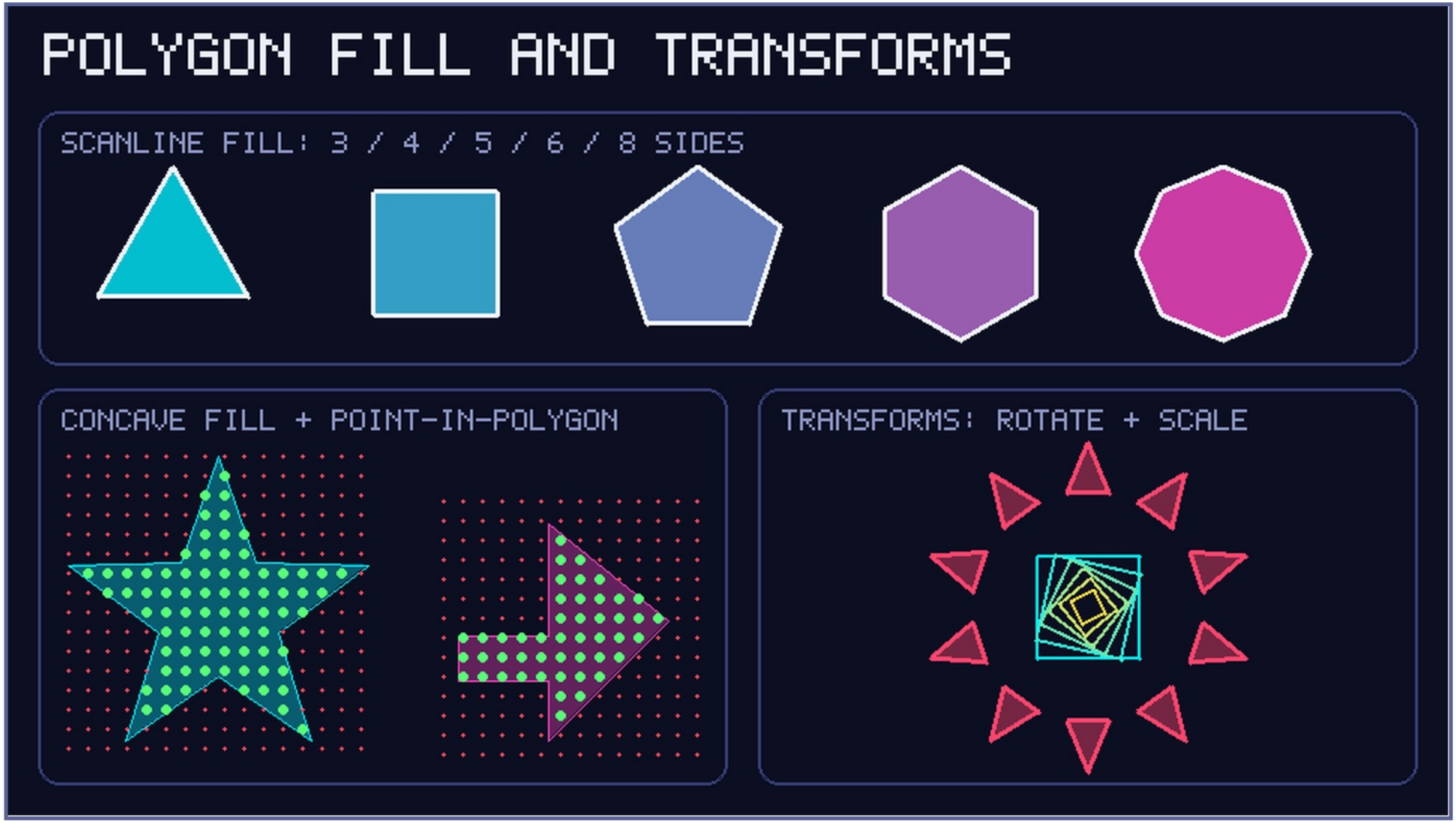
Midpoint circle, midpoint ellipse (two regions), filled versions, arcs and thick arcs.

Source: src/raster/circle.cpp, demos/demo_circles.cpp

Used in PulseRun for: jump orbs, jump pads, difficulty faces, death shock-wave rings, rounded UI corners.

02_circles.png

Polygon fill and transforms



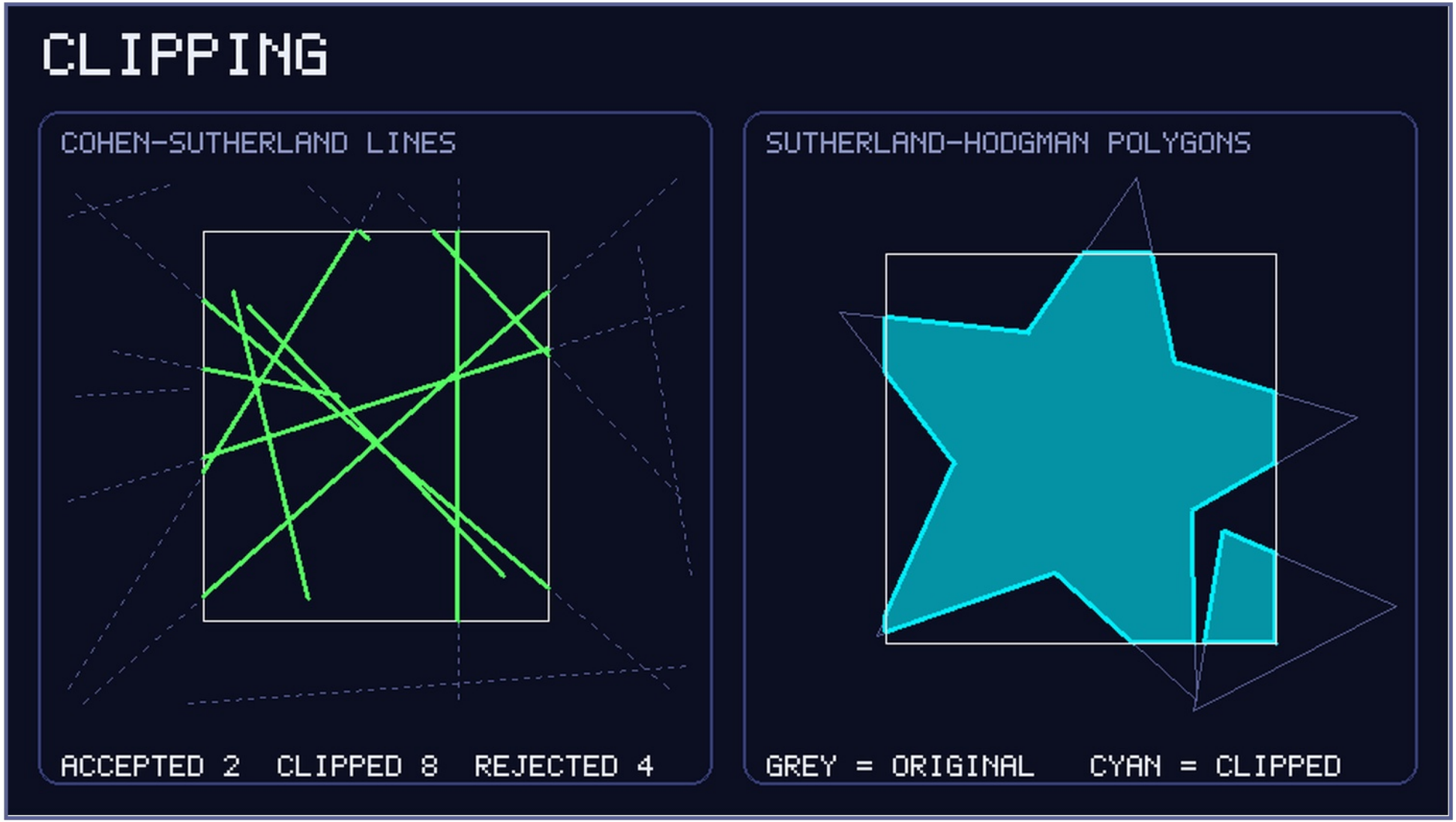
Scanline fill (edge table + active edge table, even-odd), point-in-polygon, rotate/scale/translate.

Source: src/raster/polygon.cpp, demos/demo_polygons.cpp

Used in PulseRun for: cube, spikes, blocks, saws, skyline, hills; point-in-polygon will drive spike collision.

03_polygons.png

Clipping



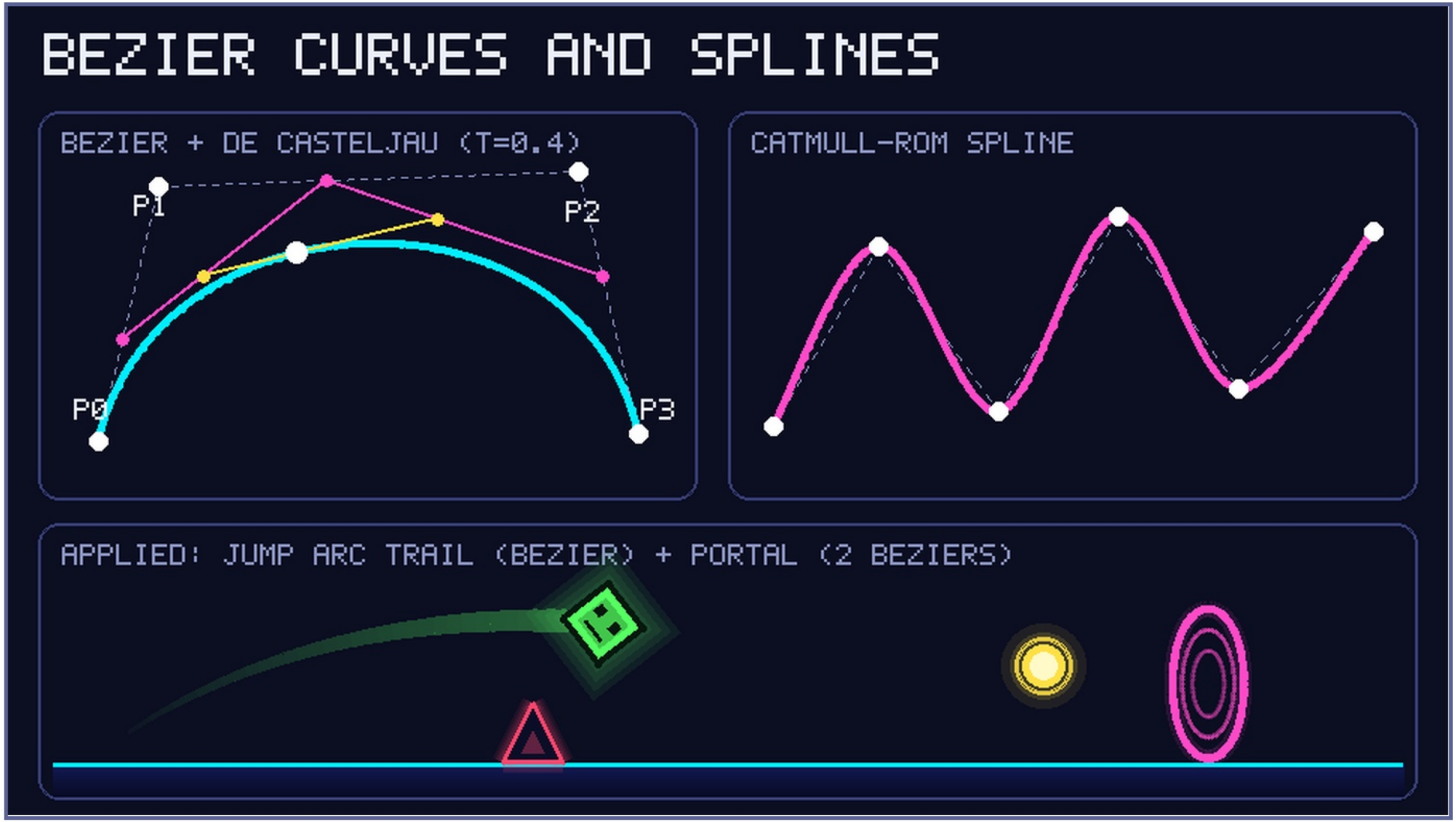
Cohen-Sutherland line clipping and Sutherland-Hodgman polygon clipping against a rectangular window.

Source: src/raster/clip.cpp, demos/demo_clipping.cpp

Used in PulseRun for: ground tick marks clipped to the ground band, death shards clipped to the viewport.

04_clipping.png

Bezier curves and splines



Cubic Bezier with the De Casteljau construction shown, Catmull-Rom spline, and the game application.

Source: src/raster/curve.cpp, demos/demo_curves.cpp

Used in PulseRun for: jump arc trail, parallax hills, portal edges, logo pulse line.

05_curves.png

Gameplay frame



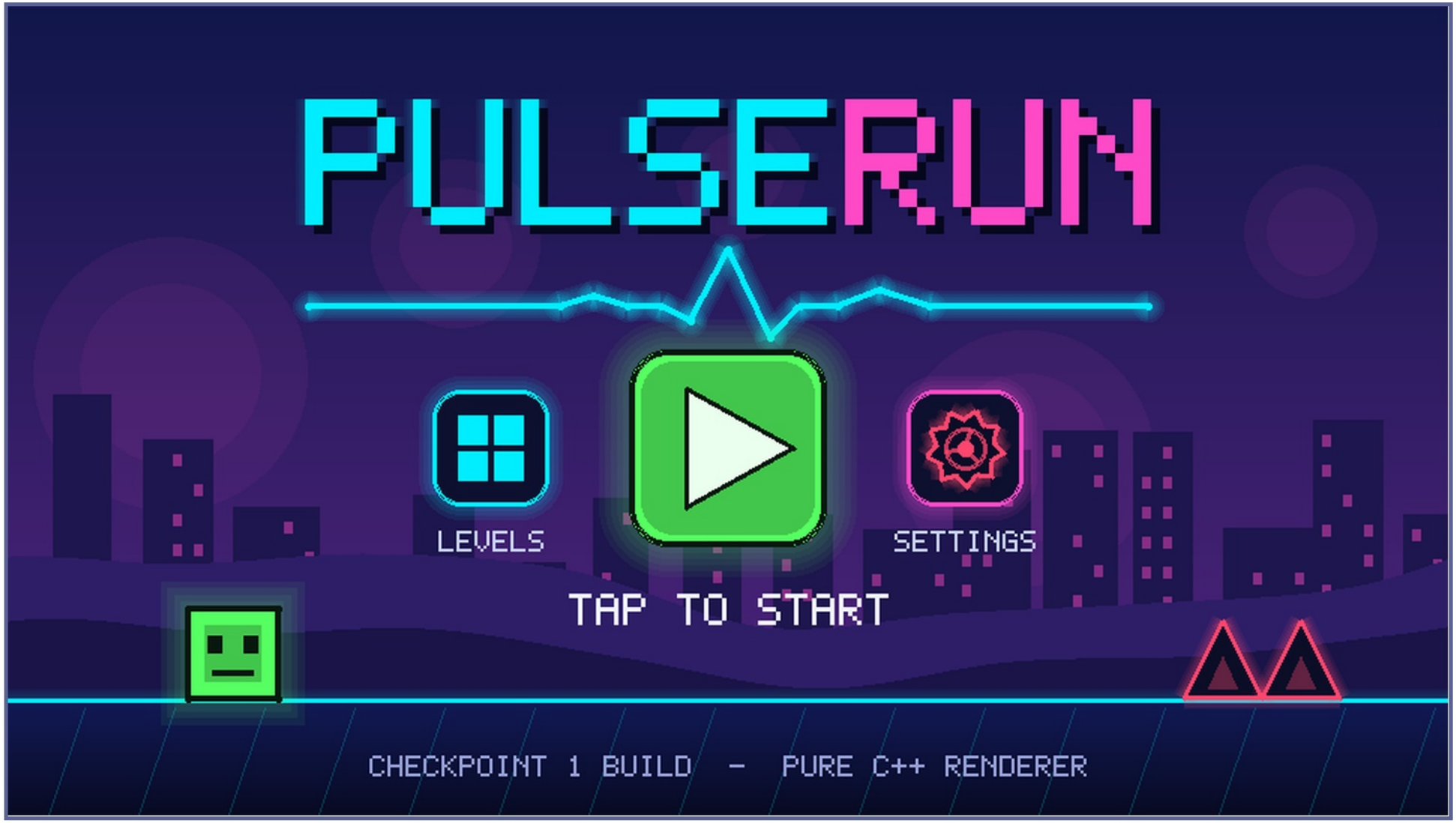
Cube mid-jump with trail, spikes, blocks, saw, orb, pad, portal, parallax city and hills, HUD.

Source: demos/demo_scenes.cpp, src/game/sprites.cpp

Every shape is drawn by the algorithms on the previous pages.

06_scene_gameplay.png

Main menu



Logo with ECG pulse line, play button, icon buttons, parallax background.

Source: demos/demo_scenes.cpp, src/ui/ui.cpp

Rounded rectangles are lines + arcs + midpoint-circle corner profiles; text uses our bitmap font.

07_scene_menu.png

Level select



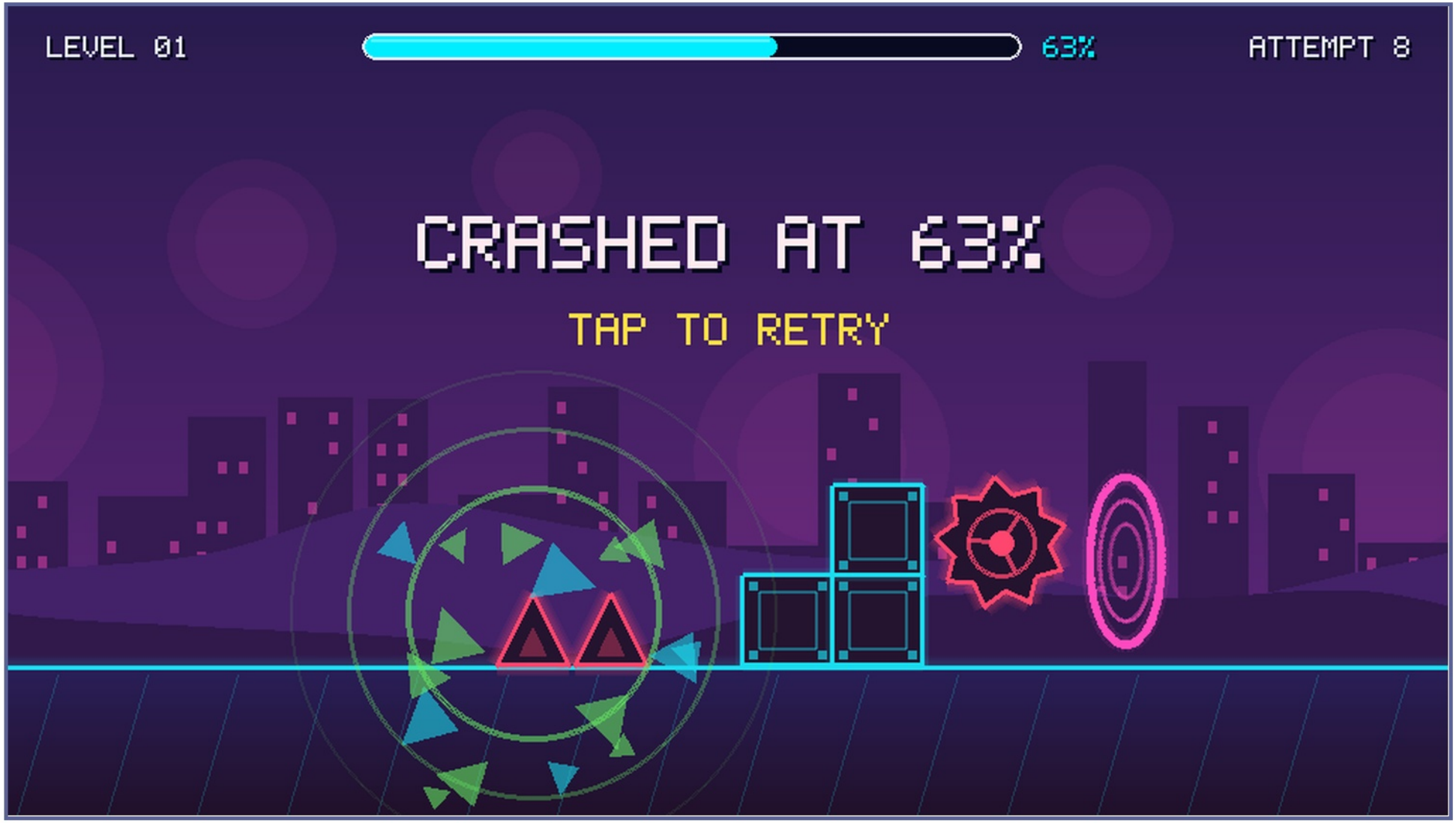
Four level cards: three unlocked with difficulty faces and best-progress bars, one locked.

Source: demos/demo_scenes.cpp, src/ui/ui.cpp

Matches the project scope: levels 1-3 committed, level 4 stretch goal.

08_scene_level_select.png

Death frame



Crash on a spike: shards flying out (clipped polygons), expanding shock-wave circles, retry prompt.

Source: demos/demo_scenes.cpp, src/game/sprites.cpp

Shards use polygon fill + Sutherland-Hodgman clipping; rings use the midpoint circle algorithm.

09_scene_death.png